

Project Documentation

Project Title

Glove Controlled Robotic Hand (Wired)

Objective

To design and develop a robotic hand that mimics human finger movements using flex sensors and servo motors controlled by an Arduino.

Hardware Used

- Arduino Mega
- 5 Flex Sensors
- 5 Servo Motors
- 10k Ω Resistors
- Breadboard
- Jumper Wires
- Glove
- Robotic Hand Structure

Methodology

1. Flex sensors are mounted on a glove.
2. The Arduino reads analog values from the sensors.
3. Sensor readings are mapped to servo motor angles.
4. Servo motors move the robotic fingers according to the user's hand movements.

Pin Connections

Component	Arduino Pin
Flex Sensor 1	A0
Flex Sensor 2	A1
Flex Sensor 3	A2
Flex Sensor 4	A3

Flex Sensor 5	A4
Servo 1	D5
Servo 2	D6
Servo 3	D7
Servo 4	D8
Servo 5	D9

Results

The robotic hand successfully replicates finger movements in real time using flex sensor inputs and servo motor actuation.

Challenges Faced

- Calibration of flex sensor values.
- Mechanical alignment of fingers.
- Stable power supply for multiple servos.

Future Scope

- Wireless control using NRF24L01 modules.
- Custom PCB development.
- Battery-powered operation.
- 3D-printed hand structure.
- Gesture recognition capabilities.

Conclusion

The project demonstrates the application of embedded systems, sensors, and actuators in creating a low-cost gesture-controlled robotic hand.